

Sample gain as a function of the study-only loss correlation

Draft note for *Crossing the Validation Crisis*

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This note derives the reference curve $G_K^{\text{test}} \approx K/(1 + (K-1)x)$, with $x = t \rho_e$, used to relate the variance-equivalent test sample gain to a study-only statistic, and explains why this statistic, rather than the redundancy score $\hat{\omega}_k^{\text{study}}$, is the quantity on which an absolute early-stopping threshold can be placed. Notation follows the paper; results taken from it are restated so that this note reads on its own, with a pointer to where they appear (section numbers of the current draft, appendix titles in full).

1 Setting and definitions

A study set of N observations is split K times by Monte-Carlo cross-validation (MCCV): each split a draws a test set $\mathcal{D}_a^{\text{te}}$ of size n_{te} uniformly without replacement, independently across splits, and trains the predictor $g_a = F_\lambda(\mathcal{D}_a^{\text{tr}}, \xi_a)$ on the remaining $n_{\text{tr}} = N - n_{\text{te}}$ observations. The test fraction is

$$t = \frac{n_{\text{te}}}{N}. \quad (1)$$

For an observation i held out by split a , let

$$d_{a,i} = \ell(g_a(x_i), y_i) - \mathcal{R}^*(g_a) \quad (2)$$

be its per-sample loss centred at the risk of the predictor that did not see it, so that $\mathbb{E}[d_{a,i} \mid g_a] = 0$, and let $\sigma_e^2 = \mathbb{E}[d_{a,i}^2]$ be the per-sample loss variance.

The benchmark-adjusted fold error of split a is $\delta_a = R_{\mathcal{D}_a^{\text{te}}}(g_a) - \hat{R}^*(g_a)$, and the K -split estimation error is $\Delta_K = \frac{1}{K} \sum_{a=1}^K \delta_a$. When the evaluation noise of the benchmark estimate $\hat{R}^*$ is negligible,

$$\delta_a = \frac{1}{n_{\text{te}}} \sum_{i \in \mathcal{D}_a^{\text{te}}} d_{a,i}. \quad (3)$$

This is the evaluation-noise approximation of the paper (Section 3, paragraph “Evaluation-noise approximation”; Appendix B, “Measuring and approximating the evaluation noise”). With $\sigma^{2,\text{te}} = \text{Var}(\delta_a)$ and $\tau^{\text{te}} = \text{Cov}(\delta_a, \delta_b)$ for $a \neq b$, the exchangeability of the splits gives

$$\text{Var}(\Delta_K) = \frac{1}{K} \sigma^{2,\text{te}} + \frac{K-1}{K} \tau^{\text{te}}. \quad (4)$$

(paper, Section 3, variance of Δ_K ; Appendix C, “Benchmark-adjusted fold errors”).

2 Step 1: the gain depends on the fold-error correlation only

The variance-equivalent test sample gain $G_K^{\text{test}} = N_K^{\text{equiv}}/n_{\text{te}}$ is the factor by which a single hold-out test set must be enlarged to match the variance of Δ_K . When the single-split variance scales as $1/m$ with the test size m , it equals the variance ratio

$$G_K^{\text{test}} \approx \frac{\text{Var}(\delta_a)}{\text{Var}(\Delta_K)} = \frac{K}{1 + (K-1) \rho_\delta}, \quad \rho_\delta = \frac{\tau^{\text{te}}}{\sigma^{2,\text{te}}} = \text{Corr}(\delta_a, \delta_b), \quad a \neq b, \quad (5)$$

where the second equality follows from (4). This is the intraclass-correlation form of the paper (Appendix D.1, “Benchmark-aware estimation of sample gain”); the gain itself is defined in Section 3 (Definitions “Variance-Equivalent Test Size” and “Test Sample Gain”). Up to the $1/m$ scaling, this step is exact: everything the gain depends on is in the fold-error correlation ρ_δ .

3 Step 2: decomposing the fold-error covariance

Single split. Given g_a , the held-out observations are i.i.d. and the $d_{a,i}$ are centred, so the cross terms within a split vanish and

$$\sigma^{2,\text{te}} = \frac{\sigma_e^2}{n_{\text{te}}}. \quad (6)$$

Two splits. For $a \neq b$, let $I_{ab} = \mathcal{D}_a^{\text{te}} \cap \mathcal{D}_b^{\text{te}}$ be the observations held out by both. Expanding the product of the two averages in (3) and separating the shared observations from the pairs of distinct observations,

$$\tau^{\text{te}} = \frac{1}{n_{\text{te}}^2} \left(\underbrace{\mathbb{E} \left[\sum_{i \in I_{ab}} d_{a,i} d_{b,i} \right]}_{\text{shared observations}} + \underbrace{\mathbb{E} \left[\sum_{i \in \mathcal{D}_a^{\text{te}}} \sum_{j \in \mathcal{D}_b^{\text{te}}, j \neq i} d_{a,i} d_{b,j} \right]}_{\text{train/test coupling}} \right). \quad (7)$$

Shared observations. Two independent uniform test sets share on average $\mathbb{E}|I_{ab}| = n_{\text{te}}^2/N = t n_{\text{te}}$ observations. On each of them the losses of the two split predictors covary by $\rho_e \sigma_e^2$, where

$$\rho_e = \frac{\mathbb{E}[d_{a,i} d_{b,i} \mid i \in I_{ab}]}{\sigma_e^2} \quad (8)$$

is the correlation of the per-sample losses of two split predictors on an observation that neither was trained on. The first term of (7) is therefore $t n_{\text{te}} \rho_e \sigma_e^2 / n_{\text{te}}^2$.

Train/test coupling. For $i \neq j$, $d_{a,i}$ and $d_{b,j}$ are not independent: i may belong to the training set of g_b , j to that of g_a , and g_a and g_b share most of their training data. This dependence is intrinsic to cross-validation and is part of τ^{te} , as the paper states when it introduces τ^{te} (Section 3; Appendix C, law of total covariance). Writing $\gamma = \mathbb{E}[d_{a,i} d_{b,j}]$ for the average covariance of such a pair, the second term is approximately γ , since there are about n_{te}^2 such pairs.

Result. Dividing by $\sigma^{2,\text{te}} = \sigma_e^2 / n_{\text{te}}$,

$$\boxed{\rho_\delta = t \rho_e + \kappa, \quad \kappa = \frac{n_{\text{te}} \gamma}{\sigma_e^2}.} \quad (9)$$

The loss correlation ρ_e measures how similarly two split predictors err on the same observation. The factor t is the share of a test set made of observations that another split also tests. The coupling contribution κ can have either sign.

4 Step 3: reference curve and study-only statistic

Substituting (9) into (5),

$$G_K^{\text{test}} \approx \frac{K}{1 + (K - 1)(t \rho_e + \kappa)}. \quad (10)$$

With $\kappa = 0$ this is the reference curve $K/(1 + (K - 1)x)$ with $x = t\rho_e$. Two limits can be read directly. For $\rho_e \rightarrow 0$, split predictors err independently and $G_K^{\text{test}} \rightarrow K$. For $\rho_e \rightarrow 1$, all split predictors err alike and $G_K^{\text{test}} \rightarrow K/(1 + (K - 1)t)$, whose limit as $K \rightarrow \infty$ is $1/t = N/n_{\text{te}}$: cross-validation then averages over the whole study set and no more.

Study-only estimate. After the first k splits of a single run, let I_{ab} be the observations held out by both splits $a < b$ and $e_{a,i} = \ell(g_a(x_i), y_i)$ the per-sample loss, computed with the loss of the metric whose gain is targeted. The pair-averaged loss covariance, loss variance and their ratio are

$$\widehat{C}_{e,k}^{\text{study}} = \binom{k}{2}^{-1} \sum_{a < b} \widehat{\text{Cov}}_{i \in I_{ab}}(e_{a,i}, e_{b,i}), \quad (11)$$

$$\widehat{V}_{e,k}^{\text{study}} = \binom{k}{2}^{-1} \sum_{a < b} \frac{1}{2} (\widehat{\text{Var}}_{i \in I_{ab}}(e_{a,i}) + \widehat{\text{Var}}_{i \in I_{ab}}(e_{b,i})), \quad (12)$$

$$\widehat{\rho}_{e,k}^{\text{study}} = \widehat{C}_{e,k}^{\text{study}} / \widehat{V}_{e,k}^{\text{study}}, \quad (13)$$

(these are the loss covariance, variance and correlation of the paper, Appendix D.3, “Study-only redundancy for early-stopping cross-validation”), and the statistic is

$$\widehat{x}_k = t \widehat{\rho}_{e,k}^{\text{study}}. \quad (14)$$

The empirical overlap $\bar{m}_k^{\text{study}}/n_{\text{te}}$, with $\bar{m}_k^{\text{study}}$ the mean of $|I_{ab}|$ over pairs, may replace t : it has the same expectation.

5 Relation to the redundancy score

The redundancy score of the paper (Appendix D.3, boxed equation; Section 5.1, paragraph “Measuring how sample gains settle in”) is $\widehat{\omega}_k^{\text{study}} = \widehat{C}_{g,k}^{\text{study}} \widehat{\rho}_{e,k}^{\text{study}} \bar{m}_k^{\text{study}}$, where $\widehat{C}_{g,k}^{\text{study}}$ is the pair-averaged covariance of the two split predictions on I_{ab} . Since $\mathbb{E} \bar{m}_k^{\text{study}} = t n_{\text{te}}$,

$$\widehat{\omega}_k^{\text{study}} \approx \widehat{C}_{g,k}^{\text{study}} n_{\text{te}} \widehat{x}_k. \quad (15)$$

The score thus contains $\widehat{x}_k$, multiplied by the prediction covariance, which carries the squared units of the target and the strength of the signal, and by the test-set size, which grows with the study. Both factors vary across settings independently of the gain. The statistic $\widehat{x}_k$ is dimensionless and bounded by t , which is why an absolute early-stopping threshold can be placed on it.

6 Assumptions and scope

1. The evaluation noise of the benchmark estimate is negligible and the single-split variance scales as $1/m$ with the test size.
2. $\widehat{\rho}_{e,k}^{\text{study}}$, computed on I_{ab} with sample centring, estimates ρ_e with a bias of order $1/|I_{ab}|$; the held-out folds must therefore contain a few observations.
3. The overlap $\mathbb{E}|I_{ab}| = t n_{\text{te}}$ holds for MCCV. For repeated K -fold, splits of the same repeat have disjoint test sets and the factor changes accordingly.
4. The reference curve omits the coupling term κ .

7 Empirical checks on simulated data

1. The study-only $\hat{\rho}_{e,k}^{\text{study}}$ matches, within 1% across all simulated designs, the population ρ_e of the same split predictors measured on the benchmarking set.
2. Varying the test fraction with everything else fixed, the fitted factor ρ_δ/ρ_e follows t (Table 1); fits with an intercept give intercepts within ± 0.01 .
3. Regression configurations and the PCam networks lie on the reference curve (PCam: median $\log(G_{20}^{\text{test}}/\text{curve})$ between -0.02 and $+0.04$). Simulated classifiers fall 10–50% below it at training sizes 100–300, that is $\kappa > 0$ there, and the gap closes by $n_{\text{tr}} \approx 800$.

Table 1: Fitted factor ρ_δ/ρ_e by test fraction (MCCV, $K = 20$, 50 seeds, training sizes 300 and 1000).

test fraction t	classification (accuracy)	regression (MSE)
0.1	0.10	0.14
0.2	0.21	0.20
0.3	0.32	0.36
0.5	0.48	0.46